

Chapter 10: The "Remodeler"

(Advanced JavaScript & DOM Lab)

I. The DOM and Grabbing Elements

- **Goal:** This chapter turns you into a "JavaScript _____," allowing your code to change the website _____ (instantly, without reloading).

The DOM (Document Object Model)

- **Definition:** When the browser loads HTML, it creates a "live," in-memory version of the page called the _____ (DOM).
- **The Split:**
 - **HTML:** The architect's _____ (the plan on paper).
 - **DOM:** The _____ building that JavaScript can remodel.
- **Rule:** JavaScript **cannot** read your .html file. It can **only** talk to the _____.

The "Grabber"

- **Command:** The magic "grabby hand" used to find and "grab" elements is _____.
- **How it Works:** It uses the _____ you use in CSS (e.g., #my-id or .my-class).

Deep Dive Discussion

1. **The Live Building:** Why is the DOM necessary? If JavaScript is the "remodeling crew," why can't it just read the index.html file directly to find the elements it needs to change?
2. **Scaffolding:** What is **scaffolding** in a professional project, and why is it important to create empty container elements (like <div class="gallery-grid"></div>) in your HTML shell before writing your JavaScript?

II. Data (Model) and the Magic Wand (Creation)

Data-Driven Design

- **Concept:** A _____ site stores its content in JavaScript, not hard-coded in HTML.
- **Arrays ([]):** A variable that acts as a _____ or list, holding multiple items in order.
- **Objects ({}):** A variable that acts as a _____, organizing related info using _____ pairs.
- **Database:** A combination of an _____ of _____ (e.g., the artworks

list).

The Magic Wand (DOM Manipulation)

These are the commands used to build and change the page structure.

Command/Property	Analogy	Description
<code>document.createElement('div')</code>	The "_____"	Creates a new HTML element invisibly in memory.
<code>container.appendChild(newElement)</code>	The "_____"	Attaches a new HTML element to the DOM, making it _____.
<code>.innerHTML</code>	The "_____"	Allows you to write a complex string of HTML and "stamp" it inside an element.
<code>.classList</code>	The "_____"	The professional way to change style by _____ or _____ a CSS class.

Application Activity

1. **Template Literal:** Rewrite the following regular string as a **Template Literal** (using backticks ``) that injects the `userName`` variable using _____.
 - `const userName = "Alex";`
 - **Regular String:** `"<h2>Hello, " + userName + "! Welcome Back.</h2>"`
 - **Template Literal:**

III. The Engine (Loops and Control)

Building with Loops

- **Loop Method:** The preferred JavaScript method for looping through every item in an Array is for...of.
- **Dynamic Construction:** Inside this loop, we use the data from the Array and the innerHTML (from section II) to build a new HTML card for each item automatically.

The Controller

- **querySelectorAll():** The "Grab _____" tool. It returns a _____ of every element that matches the selector.
- **Making it Interactive:** To add a click listener to all the dynamically created elements, you must:
 1. Use _____ to grab all the elements.
 2. Use a _____ loop to iterate over that list.
 3. Add an _____ to **each** element.

Deep Dive Discussion

1. **Efficiency of Data-Driven Design:** Once the Cosmic Canvas site is launched, an artist wants to add a new piece of art. What is the **only thing** they need to change in the entire project to make that new art appear on the live site?
2. **Model-View-Controller (MVC):** This chapter introduces the MVC pattern. Match the following elements to their correct role:
 - **Model:** _____ (Role: The Data)
 - **View:** _____ (Role: The HTML/CSS Shell)
 - **Controller:** _____ (Role: The JavaScript Logic/Event Listeners)

Application Activity

1. **DOM Grab and Change:** Write the JavaScript code that first **grabs** the HTML element with the ID main-title and then **changes its visible content** to "Welcome to the Live DOM!"
 - **Code:**

```
// Grab the element:
```

```
const titleElement =
```

[illegible]

```
// Change its content:
```

```
titleElement.\ \ \ \ \ \ \ \ \ \ \ \ \ \ \ = "Welcome to the Live DOM!";
```